

Q18 Ans: D

The resistors are connected in parallel to the power supply.

$$R_{effective} = \left(\frac{1}{1300} + \frac{1}{1900} \right)^{-1} = 772 \, \Omega$$

$$I = \frac{240}{772} = 0.31 \, \text{A}$$

$$P_{total} = \frac{240^2}{1300} + \frac{240^2}{1900} = 75 \, \text{W}$$

Q18 Ans: C